

Marketplace Analytics Platform

Business Intelligence • Forecasting • Executive Reporting

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Marketplace Analytics Platform

Business Intelligence • SQL Database • Forecasting • Executive Dashboard

Portfolio case study using synthetic ecommerce data

TOTAL MARKETPLACE GMV

\$5.33M

TOTAL GROSS PROFIT

\$2.92M

TRANSACTIONS

23,016

Built to show business analysis, dashboard storytelling, and predictive analytics in one clean project.

Project Story

What this project is solving

- Simulate a product-based marketplace with sales, inventory, customers, calendar factors, search trends, and competitors.
- Build an end-to-end analytics workflow: data model → SQL database → Python analysis → dashboard → forecasting.
- Turn raw transactional data into business insights around GMV, category performance, seasonality, and operational planning.

Business framing

Which products/categories drive revenue?

Where does seasonality impact demand?

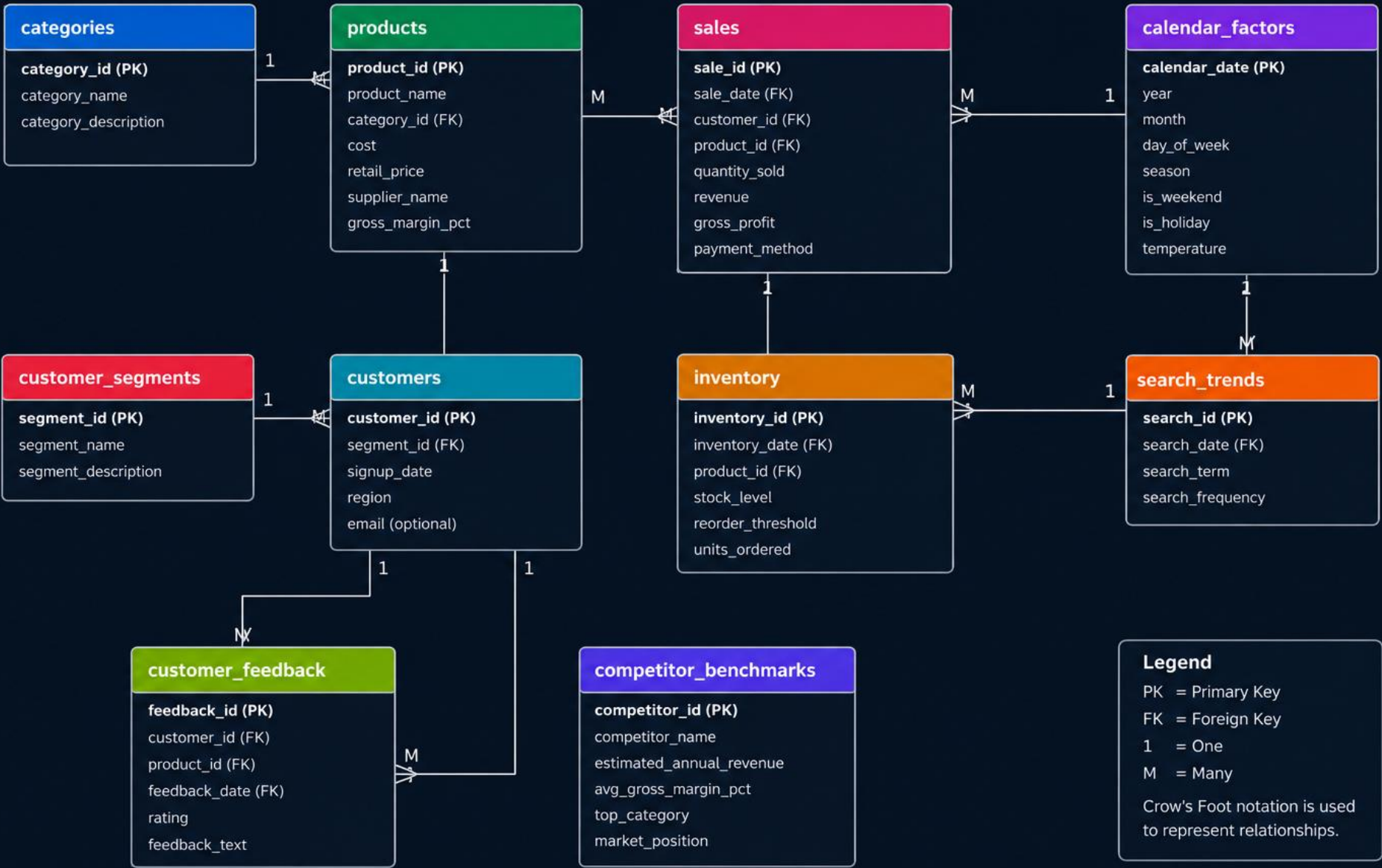
How can forecasting support inventory decisions?

Data Model & ER Diagram

Relational structure behind the analytics workflow

Marketplace Analytics Platform — Product-Based ER Diagram

Clean relational model for product sales, customer segments, inventory, calendar factors, search trends, and market benchmarks.



Executive Dashboard Screenshot

Dashboard view used for GitHub and presentation storytelling

MARKETPLACE EXECUTIVE DASHBOARD

Overview of key performance metrics and revenue insights

Select Date Range

Jan 2023 - Dec 2023



TOTAL MARKETPLACE GMV

\$5.33M



TOTAL GROSS PROFIT

\$2.92M



TRANSACTION VOLUME KPI

23,016



MARKETPLACE MONTHLY REVENUE PERFORMANCE



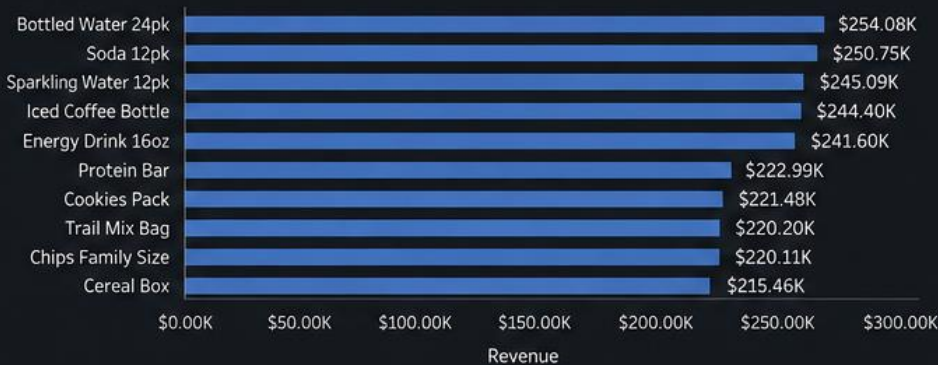
MARKETPLACE REVENUE BY SEASON



MARKETPLACE REVENUE BY PRODUCT CATEGORY



TOP 10 PRODUCTS BY MARKETPLACE REVENUE



Note: All values are in USD. K = Thousands, M = Millions

Core KPI Summary

High-level marketplace performance

TOTAL MARKETPLACE GMV

\$5.33M

TOTAL GROSS PROFIT

\$2.92M

GROSS MARGIN

54.79%

TRANSACTIONS

23,016

CUSTOMERS

2,500

AVG TRANSACTION

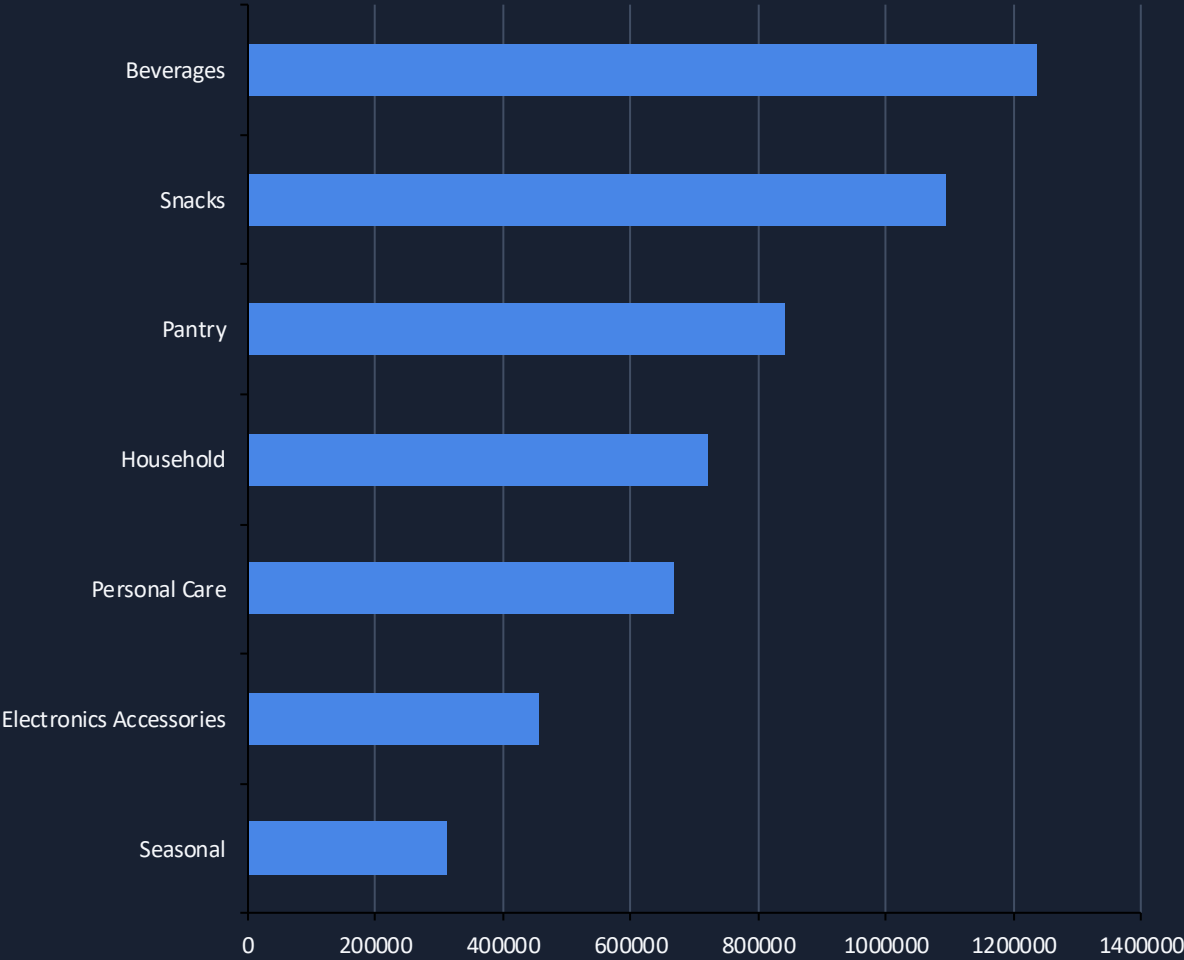
\$231.57

The project shows a multi-year synthetic marketplace with enough volume to support KPI tracking and trend analysis. GMV and gross profit provide the executive view; transaction volume and average transaction value explain operational activity. This KPI layer becomes the foundation for dashboard reporting, Excel summaries, and interview talking points.

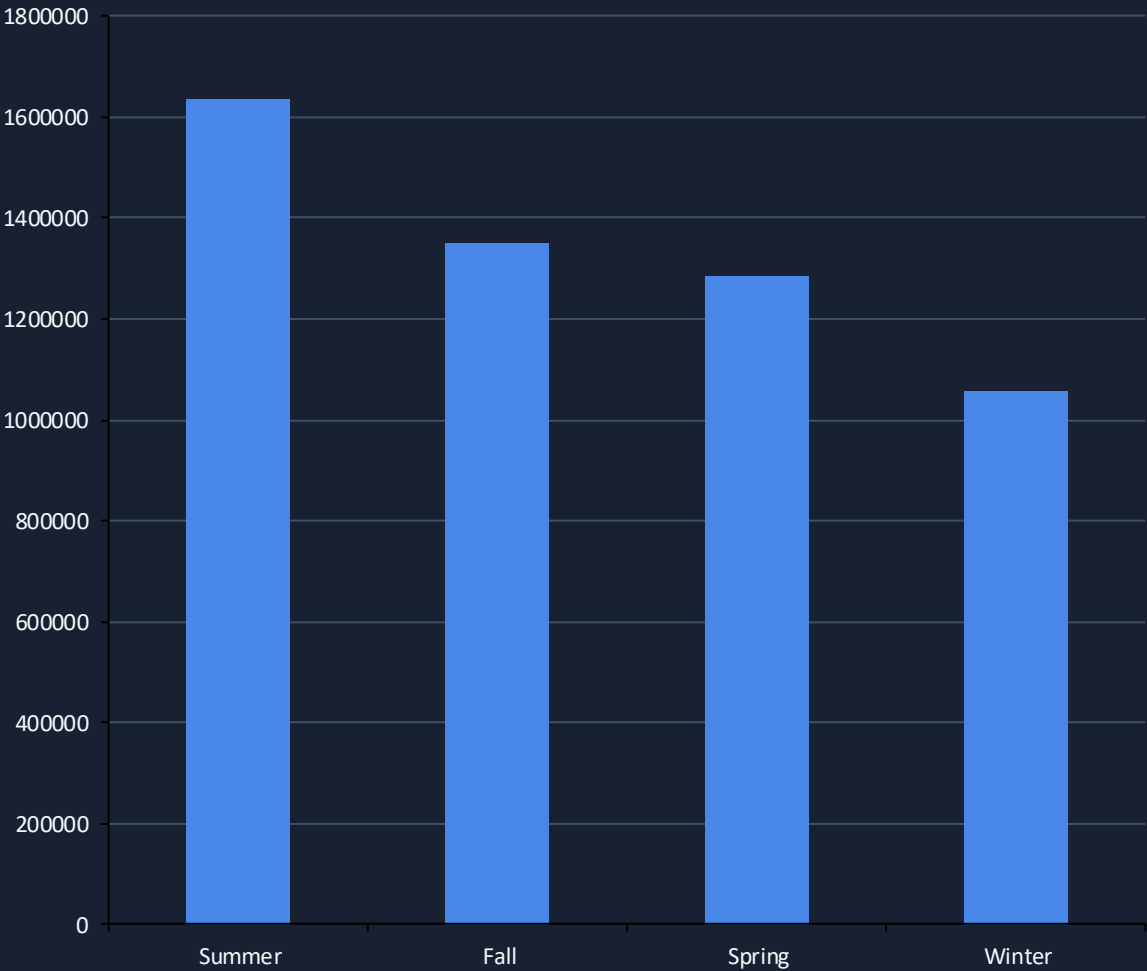
Revenue Drivers

Category performance and seasonal movement

Revenue by Product Category



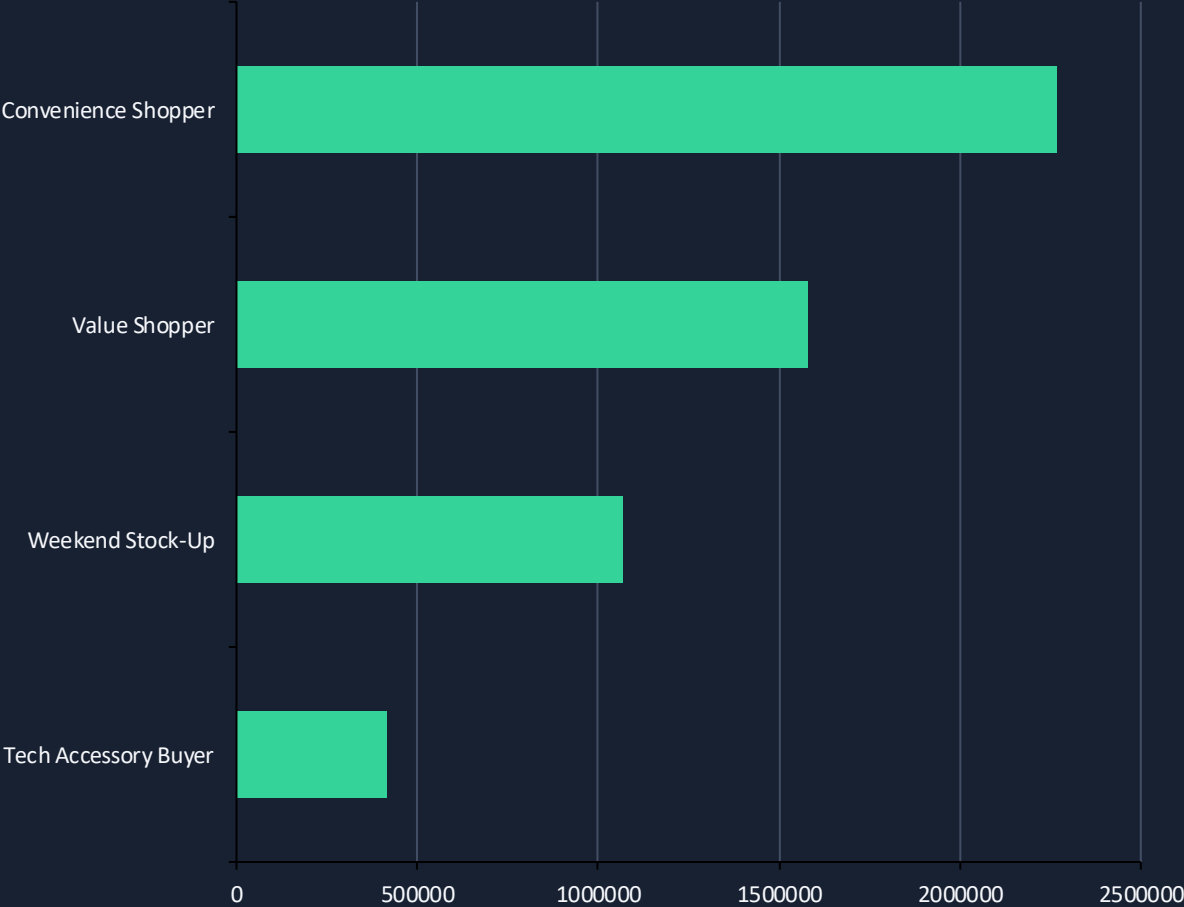
Revenue by Season



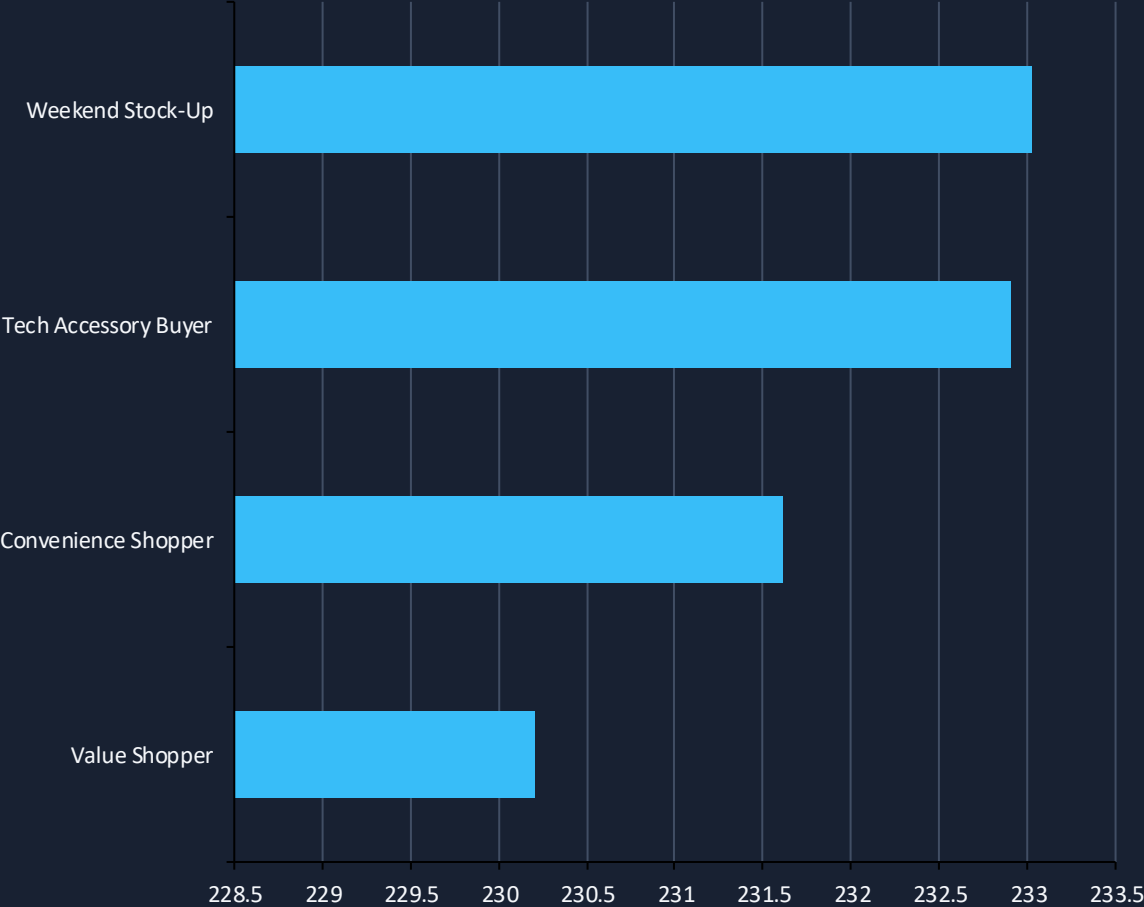
Customer Segment Insights

Understanding who contributes to GMV

Revenue by Customer Segment



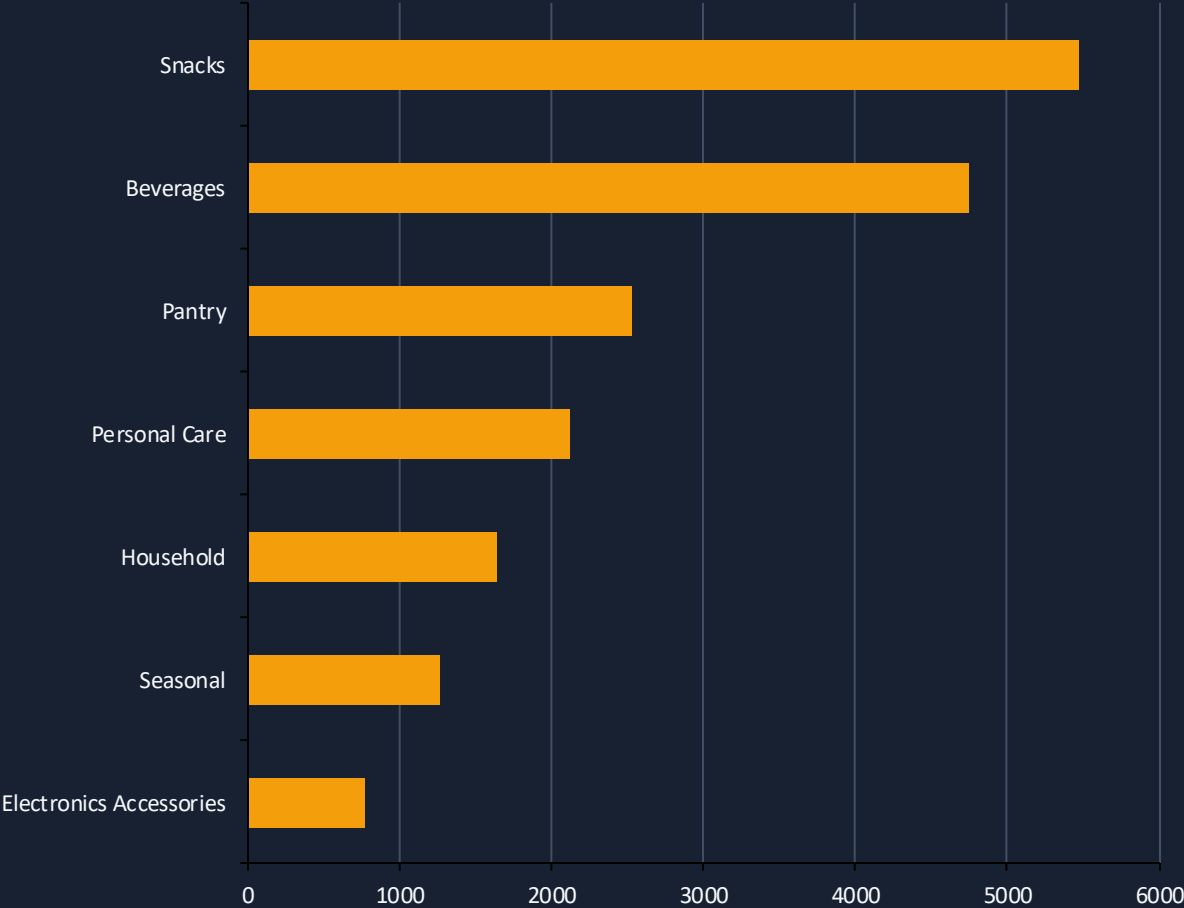
Avg Transaction Value by Segment



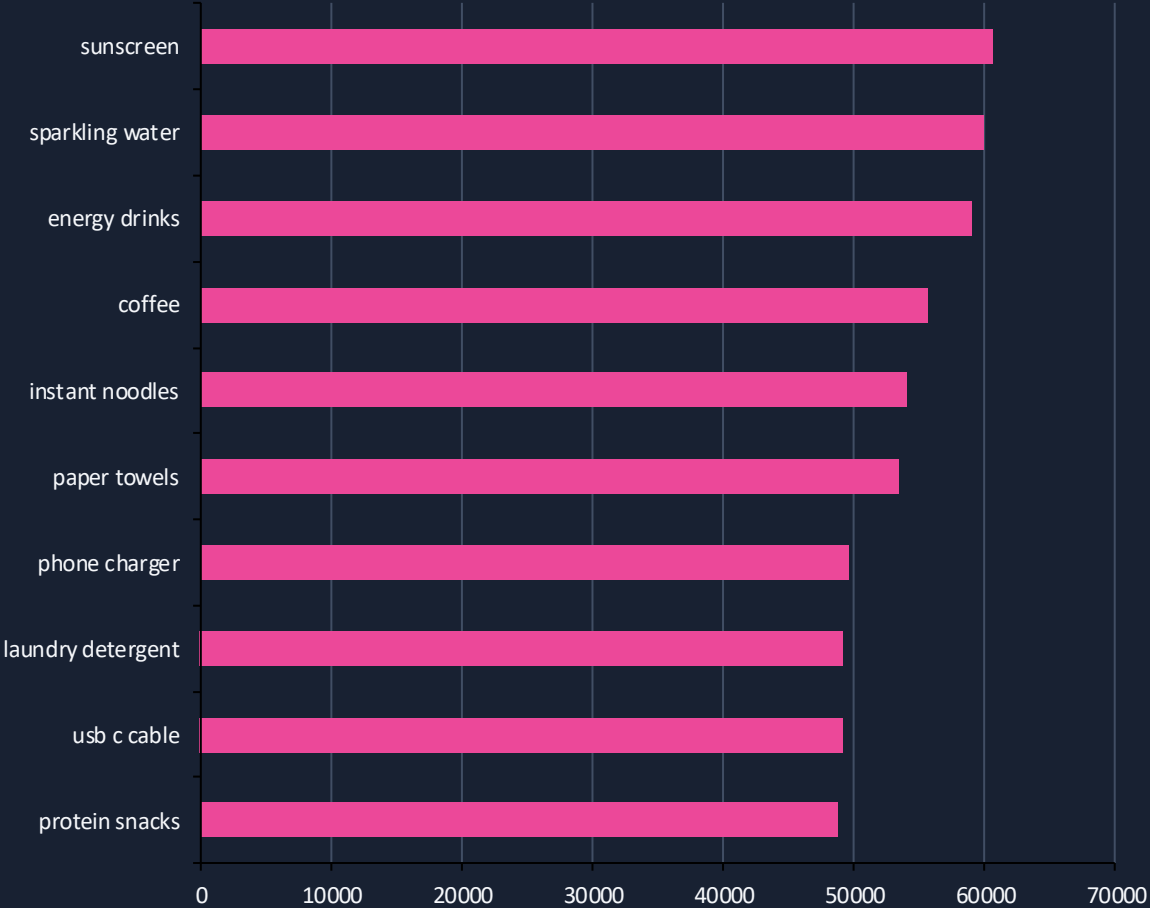
Operations & Demand Signals

Inventory and search behavior supporting planning

Units Ordered by Category



Top Search Demand Signals



Forecasting & Predictive Analytics

Model comparison and feature importance

Model Performance

Linear Regression R^2 : 0.746

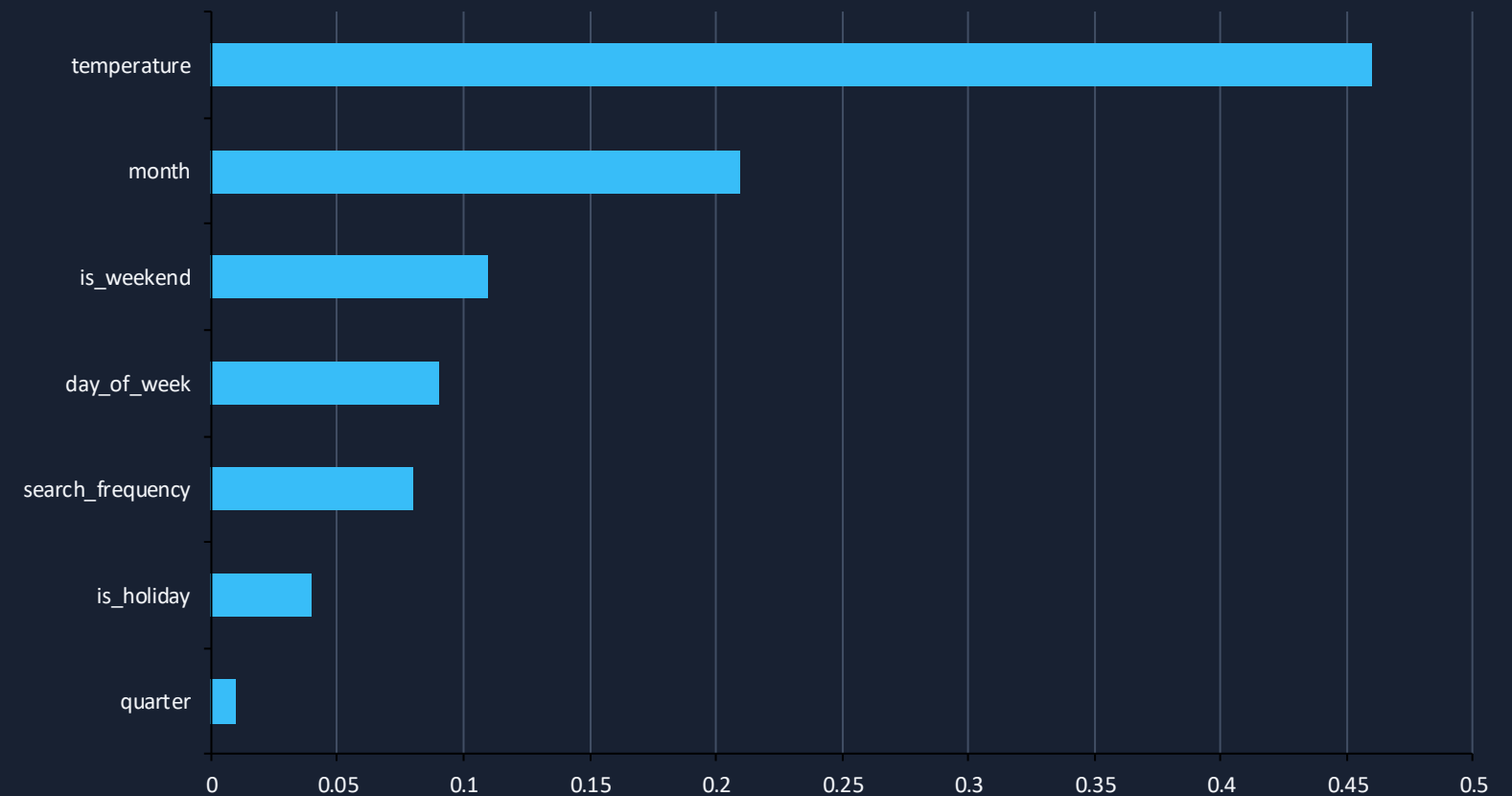
Random Forest R^2 : 0.832

Random Forest MAE: \$309.01

Random Forest RMSE: \$392.40

Takeaway: Random Forest performed better because it captured nonlinear demand patterns more effectively.

Random Forest Feature Importance



SQL Database & GitHub Deliverables

How the project is packaged

Database Layer

PostgreSQL relational database
create_tables.sql for schema setup
marketplace_sql_queries.sql for analytics queries
Joins across sales, products, categories, customers, and
calendar factors

Why it matters

Shows SQL fluency
Shows data modeling
Makes the project reproducible

Repo Assets

Python notebooks
SQL scripts
CSV datasets
ER diagram
Dashboard screenshots
Excel workbook
PowerPoint presentation
README documentation

How I Would Explain This in an Interview

Simple, honest, and business-focused

- I built a synthetic marketplace analytics platform to simulate ecommerce strategy and operations analysis.
- I started by designing a relational data model, then loaded the data into PostgreSQL and wrote SQL queries for KPIs and category performance.
- I used Python to validate the data, calculate business metrics, analyze seasonality, and build forecasting models.
- I then created dashboards and Excel reporting outputs to turn the analysis into something executive-friendly.
- The main insight was that revenue was driven by beverages/snacks, summer demand, and weekend purchasing patterns.

My focus was connecting data → insight → recommendation.

Final Summary

What this project demonstrates

- End-to-end analytics workflow from data model to executive reporting.
- SQL database design and analytics query development.
- Python-based KPI analysis, segmentation, seasonality, and forecasting.
- Dashboard reporting for marketplace performance monitoring.
- Business recommendations tied to inventory planning, seasonality, and demand signals.